

PROJECT REPORT

Superstore Sales Analytics, Profit Prediction and Customer Segmentation Using Machine Learning

AICTE | IBM SkillsBuild Data Analytics with AI Internship 2026

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Project Type	Data Analytics and Machine Learning
Dataset	Superstore Retail Dataset
Primary Notebook	Saini_Venkatraman_Superstore_Analytics.ipynb

This report documents the completed analysis, preprocessing, exploratory data analysis, statistical analysis, machine-learning models, evaluation results, customer segmentation, outputs, limitations and future enhancements implemented in the project notebook.

Table of Contents

1. Abstract
2. Introduction
3. Problem Statement
4. Objectives
5. Dataset Description
6. Tools and Technologies
7. Project Methodology
8. Data Loading and Quality Analysis
9. Exploratory Data Analysis
10. Feature Engineering
11. Statistical Analysis
12. Machine Learning and Predictive Analytics
13. Regression - Sales Prediction
14. Classification - Profitability Prediction
15. Customer Segmentation Using K-Means
16. Model Evaluation Summary
17. Key Findings and Business Interpretation
18. Output Files
19. Limitations
20. Future Enhancements
21. Conclusion
22. References
23. Project Structure
24. Declaration

1. Abstract

This project applies data analytics and machine-learning techniques to the Superstore retail dataset to understand sales, profit, customer purchasing behavior, product performance and regional patterns. The workflow covers data loading, inspection, quality checking, exploratory data analysis, feature engineering, statistical analysis, supervised machine learning and customer segmentation.

Three machine-learning approaches were implemented. Linear Regression was used for Sales prediction, Decision Tree Classification was used to classify orders as Profitable or Non-Profitable, and K-Means clustering was used to segment customers according to purchasing behavior. The regression model achieved an R^2 of 0.1862, while the profitability classifier achieved 90.15% accuracy and an F1 score of 0.9355. K-Means evaluation across K=2 to K=6 selected two customer clusters based on the highest observed silhouette score of 0.3672.

2. Introduction

Retail organizations generate transactional data containing information about customers, products, sales, discounts, shipping and profitability. Data analytics can transform this raw information into structured insights that support business understanding and decision-making.

The project analyzes historical Superstore data using Python-based analytics tools. Exploratory analysis is used to identify patterns in sales and profit, while machine learning extends the analysis into prediction and customer segmentation.

3. Problem Statement

The objective is to analyze historical Superstore sales data and identify patterns related to sales, profit, customers, products and regions. The project also aims to use machine-learning techniques to predict sales-related outcomes, classify orders based on profitability, and identify groups of customers with similar purchasing behavior.

4. Objectives

- Understand and preprocess the Superstore dataset.
- Check and handle data-quality issues such as missing values and duplicate records.
- Perform exploratory data analysis on sales, profit, customers, products and regions.
- Analyze trends using statistical summaries and visualizations.
- Create useful features from the existing dataset.
- Build a regression model for sales prediction.
- Build a classification model for profitability prediction.
- Apply K-Means clustering for customer segmentation.
- Evaluate models using appropriate performance metrics.
- Export important machine-learning outputs for reuse and reporting.

5. Dataset Description

The analysis uses the Superstore retail dataset supplied for the project. The Orders dataset contains 9,994 records and 21 columns. Additional files include Returns with 296 records and People with 4 records.

Dataset component	Rows	Columns
Orders	9,994	21
Returns	296	2
People	4	2

The Orders data contains fields covering order and shipping dates, shipping mode, customer information, segment, location, product category and sub-category, sales, quantity, discount and profit.

Dataset source/link for submission: [Kaggle - Superstore Dataset](#)

6. Tools and Technologies

Technology / Library	Purpose
Python	Primary programming language
Jupyter Notebook	Interactive project development and documentation
Pandas	Data loading, cleaning, transformation and aggregation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Scikit-learn	Regression, classification, preprocessing, scaling and clustering
K-Means	Customer segmentation algorithm
Linear Regression	Sales prediction model
Decision Tree Classifier	Profitability classification model

7. Project Methodology

The project follows a structured analytics and machine-learning workflow:

Raw Dataset → Data Loading → Inspection → Data Quality Checking → EDA → Statistical Analysis → Feature Engineering → Regression / Classification / Clustering → Evaluation → Result Export → Business Interpretation

For supervised learning, the dataset was divided into training and testing sets. The training portion contained 7,995 records and the testing portion contained 1,999 records.

8. Data Loading and Quality Analysis

8.1 Data inspection

The Orders dataset was loaded and inspected for dimensions, columns, data types, descriptive statistics, missing values and duplicates. Order Date and Ship Date were converted to datetime data types.

8.2 Data quality

The notebook checked missing values and duplicate records. The completed analysis reported zero missing values across the Orders columns and zero duplicate rows. This provided a clean base for subsequent analysis.

8.3 Descriptive statistics

Measure	Mean	Minimum	Maximum
Sales	229.858	0.444	22,638.480
Quantity	3.790	1	14
Discount	0.156	0.000	0.800
Profit	28.657	-6,599.978	8,399.976

9. Exploratory Data Analysis

9.1 Overall KPIs

KPI	Value
Total Sales	\$2,297,200.86
Total Profit	\$286,397.02
Total Quantity	37,873
Total Orders	5,009
Total Customers	793
Total Products	1,862
Average Order Value	\$458.61

9.2 Category analysis

Category	Sales	Profit	Quantity
Technology	\$836,154.03	\$145,454.95	6,939
Furniture	\$741,999.80	\$18,451.27	8,028
Office Supplies	\$719,047.03	\$122,490.80	22,906

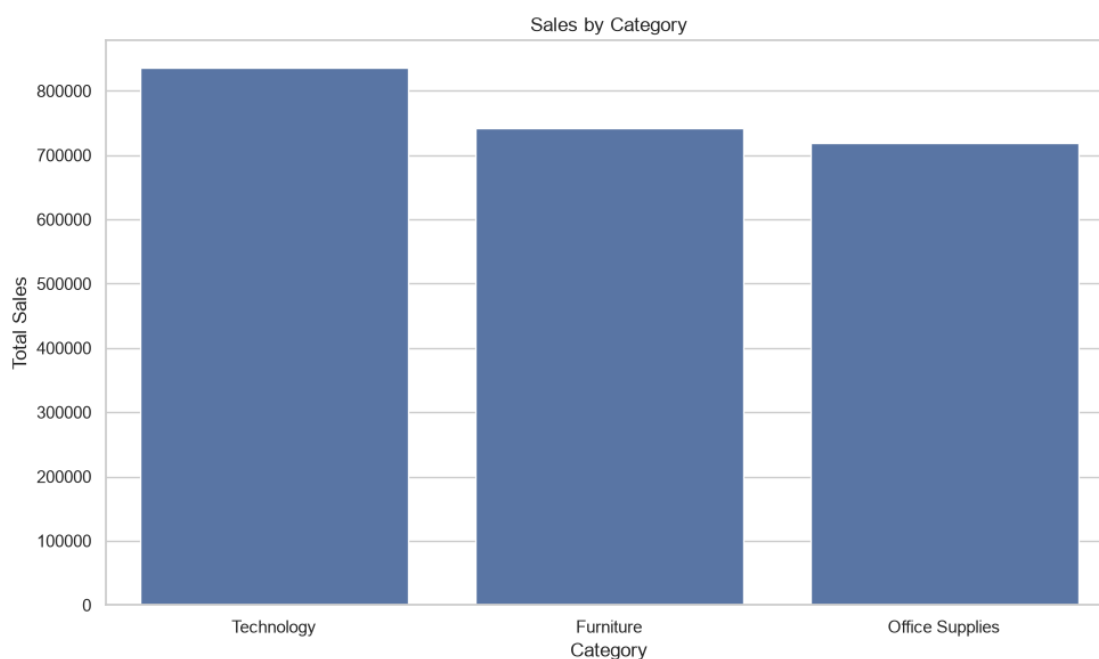


Figure: Category-level sales and profit visualization generated in the notebook.

9.3 Yearly sales trend

Year	Sales
2014	\$484,247.50
2015	\$470,532.51
2016	\$609,205.60
2017	\$733,215.26

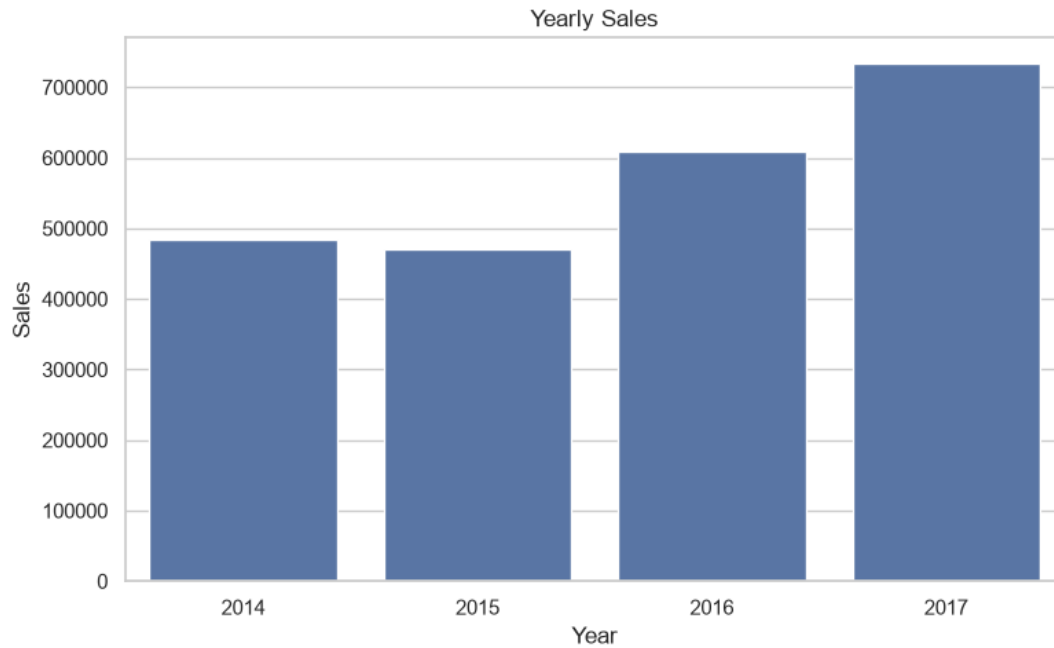


Figure: Year-wise sales visualization generated in the notebook.

9.4 Regional performance

Region	Sales	Profit
West	\$725,457.82	\$108,418.45
East	\$678,781.24	\$91,522.78
Central	\$501,239.89	\$39,706.36
South	\$391,721.91	\$46,749.43

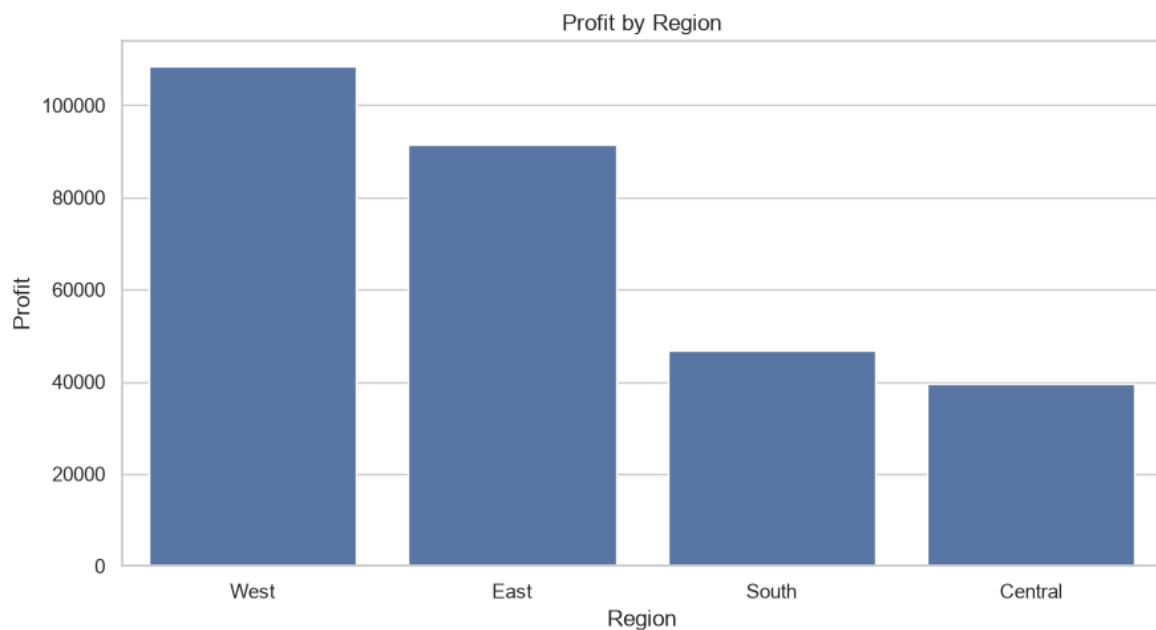


Figure: Regional sales/profit visualization generated in the notebook.

9.5 Sub-category observations

Phones recorded the highest sales among the listed sub-categories at approximately \$330,007.05, followed closely by Chairs at approximately \$328,449.10. Tables recorded approximately \$20,965.53 in sales in the notebook output and a negative profit of approximately \$17,725.48.

10. Feature Engineering

The notebook created additional variables to support temporal, operational and profitability analysis:

Feature	Description
Year	Year extracted from Order Date
Month	Month extracted from Order Date
Month Name	Month name derived from Order Date
Quarter	Quarter derived from Order Date
Shipping Days	Difference between Ship Date and Order Date
Profit Margin	Profit divided by Sales, expressed as a percentage with zero handling
Profitability	Profitable when Profit > 0; otherwise Non-Profitable

11. Statistical Analysis

The project used descriptive statistics and correlation analysis to understand numerical relationships. The notebook reported a correlation of -0.2195 between Discount and Profit, indicating a negative linear association in this dataset. Correlation describes association and does not by itself establish causation.

The profitability distribution contained 8,058 Profitable orders and 1,936 Non-Profitable orders, corresponding to 80.63% and 19.37% respectively.

12. Machine Learning and Predictive Analytics

Three machine-learning approaches were implemented: regression for Sales prediction, classification for order profitability, and clustering for customer segmentation. Supervised models used training and testing subsets, while clustering operated on aggregated customer-level features.

13. Regression - Sales Prediction

A Linear Regression model was developed to predict Sales. Numerical features were Quantity, Discount and Shipping Days. Categorical features were Ship Mode, Segment, Region, Category and Sub-Category. Numerical preprocessing used median imputation, while categorical preprocessing used most-frequent imputation and one-hot encoding.

Metric	Result
MAE	226.31
MSE	480,704.47
RMSE	693.33
R ²	0.1862

The R² value indicates that the selected linear model explains a limited portion of the variation in Sales. The result is therefore useful as a documented baseline model, while leaving room for feature and algorithm improvements.

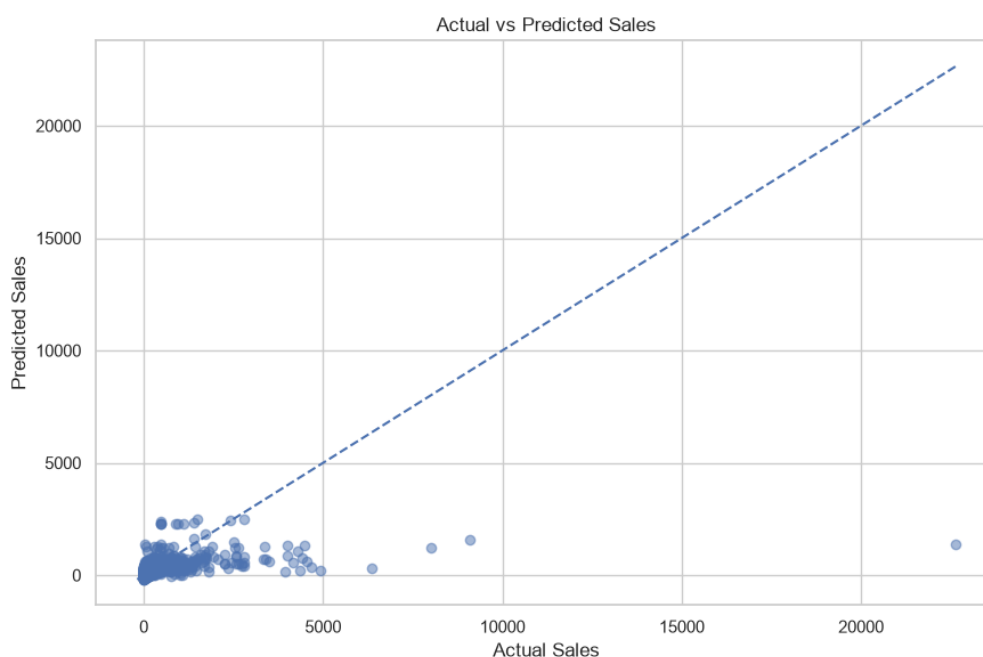


Figure: Actual versus predicted Sales visualization generated by the notebook.

14. Classification - Profitability Prediction

A Decision Tree Classifier was used to classify orders as Profitable or Non-Profitable. The model used Sales, Quantity, Discount, Shipping Days and categorical features including Ship Mode, Segment, Region, Category and Sub-Category. The classifier used max_depth=6, random_state=42 and balanced class weights.

Metric	Result
Accuracy	0.9015
Precision	0.9910

Metric	Result
Recall	0.8859
F1 Score	0.9355

The test set contained 1,999 records. The classification report showed different performance across the two classes, so the overall accuracy should be interpreted together with precision, recall and F1 score.

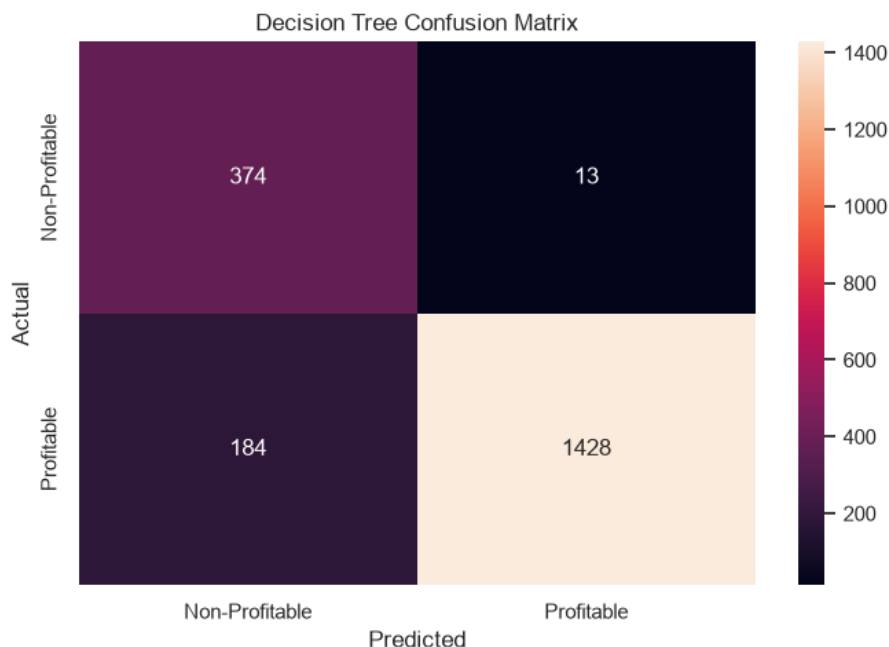


Figure: Confusion matrix and classification visualization generated by the notebook.

15. Customer Segmentation Using K-Means

Customer-level aggregation was created using Total Sales, Total Profit, Total Quantity, Order Count and Average Order Value. These variables were standardized using StandardScaler before applying K-Means clustering.

K	Silhouette Score
2	0.3672
3	0.3487
4	0.3215
5	0.3087
6	0.3140

K=2 was selected based on the highest silhouette score observed among the tested values. The resulting customer segments were then profiled using their average purchasing and profitability measures.

Cluster	Customers	Avg Sales	Avg Profit	Avg Quantity	Avg Orders	Avg Order Value
0	290	\$5,142.92	\$788.45	70.82	8.51	\$314.77
1	503	\$1,601.90	\$114.80	34.47	5.05	\$177.76

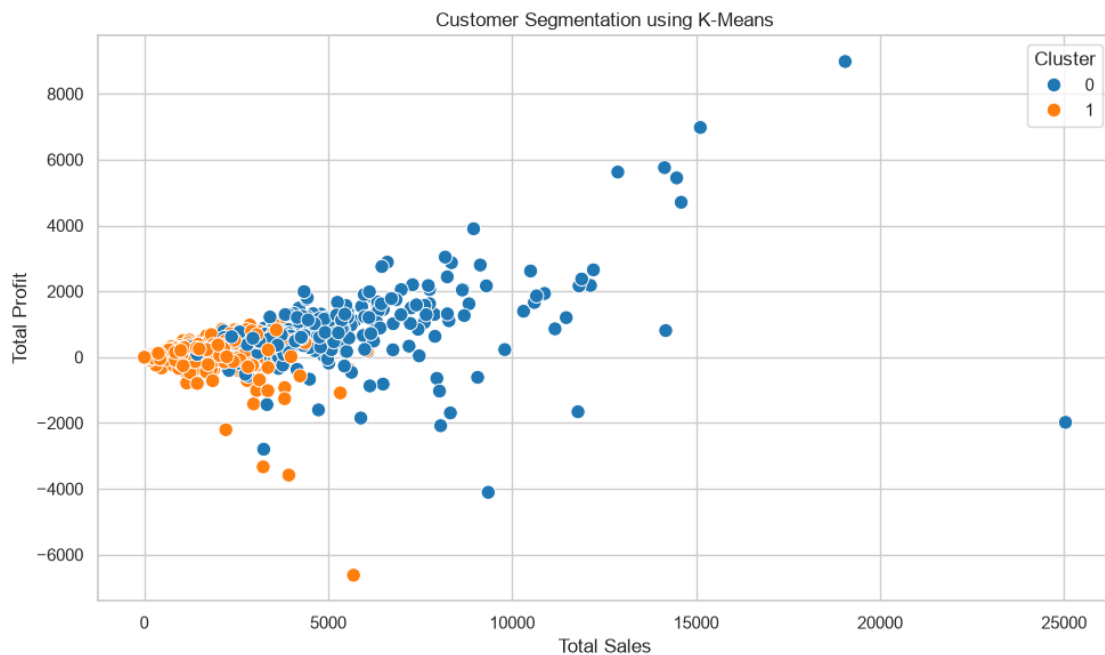


Figure: Customer segmentation scatter visualization generated by the notebook.

16. Model Evaluation Summary

Task	Algorithm	Main Result
Sales Prediction	Linear Regression	$R^2 = 0.1862$; RMSE = 693.33
Profitability Classification	Decision Tree	Accuracy = 0.9015; F1 = 0.9355
Customer Segmentation	K-Means	K=2; silhouette = 0.3672

17. Key Findings and Business Interpretation

- The dataset contains 9,994 order-level records and supports analysis across products, customers, regions, shipping and profitability.
- Technology generated the highest sales and profit among the three major categories in the notebook results.
- Sales increased across the four observed years, with 2017 recording the highest annual sales value in the analysis.
- The West region recorded the highest sales and profit among the four regions.
- The notebook identified a negative correlation of -0.2195 between Discount and Profit.
- The profitability classifier achieved 90.15% accuracy on the test set, with an F1 score of 0.9355.
- Customer clustering produced two segments with different average sales, profit, order frequency and order value profiles.
- The Linear Regression model provides a baseline but its R^2 of 0.1862 indicates that additional features or alternative models may be needed for stronger Sales prediction.

18. Output Files

File	Purpose
regression_results.csv	Regression evaluation metrics

File	Purpose
classification_results.csv	Classification evaluation metrics
customer_segments.csv	Customer-level records with assigned cluster
cluster_profile.csv	Aggregated profile of customer clusters

19. Limitations

- The analysis is based on historical Superstore data and therefore describes the patterns present in that dataset.
- The Linear Regression model achieved a relatively limited R^2 of 0.1862.
- K-Means segmentation depends on the selected customer features, standardization procedure and tested values of K.
- The project focuses on the models implemented in the notebook and does not include extensive hyperparameter optimization or deployment of the models as a production service.

20. Future Enhancements

- Compare additional regression algorithms and tune model hyperparameters.
- Experiment with additional classification algorithms and model-selection techniques.
- Extend customer segmentation with additional behavioral or monetary features.
- Build an interactive dashboard using Tableau, Power BI or Streamlit.
- Deploy the analytical application or prediction service for interactive use.
- Add automated model evaluation and monitoring for future datasets.

21. Conclusion

The project demonstrates an end-to-end data analytics and machine-learning workflow using the Superstore retail dataset. It combines data preparation, exploratory analysis, visualization, statistical analysis, feature engineering, regression, classification and customer segmentation within a single documented Jupyter Notebook.

The completed analysis provides quantitative measures of sales, profit, regional and category performance, while the machine-learning components demonstrate three different analytical perspectives: predicting Sales, classifying order profitability, and identifying customer groups. The exported CSV files provide reusable outputs for further reporting or dashboard development.

22. References

1. Superstore dataset source: [Kaggle - Superstore Dataset](#)
2. Project implementation and results: Saini_Venkatraman_Superstore_Analytics.ipynb
3. Python data-analysis and machine-learning libraries used in the project: Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn.

23. Project Structure

Component	Description
Saini_Venkatraman_Superstore_Analytics.ipynb	Complete analysis notebook
sample_-_superstore.xls	Input Superstore dataset
regression_results.csv	Regression results
classification_results.csv	Classification results
customer_segments.csv	Customer segmentation output
cluster_profile.csv	Cluster summary
README.md	Project overview, technologies, setup and usage instructions
requirements.txt	Python package dependencies

24. Declaration

I hereby declare that this project report documents the analysis and implementation completed as part of the AICTE | IBM SkillsBuild Data Analytics with AI Internship 2026. The report is based on the submitted project notebook and its generated analytical results.

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